Comparison of automated methods for AI Video Detection

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ABSTRACT

As generative AI tools make realistic video synthesis increasingly accessible, reliably detecting AI-generated video has become a critical challenge. We benchmark multiple CNN-based architectures on the IVY-Fake dataset under a strict open-set protocol, finding that a LCRN achieves the best performance at 77.80% accuracy and AUC of 0.9186. A consistent validation-to-test gap across all models points to source-signature learning as the key barrier to generalisation, with training data diversity identified as the primary bottleneck.

RESULTS

All architectures achieve high validation accuracy but show a consistent drop at test time, with accuracy ranging from 70% to 78% across models. The LCRN performs strongest overall, achieving 77.80% test accuracy and an AUC of 0.9186, followed closely by the Video Swin Transformer.

Model	Val Acc	Test Acc	F1	Precision	Recall	AUC
LCRN	0.9825	0.7780	0.7181	0.9774	0.5675	0.9186
ReStraV	0.8461	0.6999	0.6444	0.7867	0.5458	0.7531
VideoSwinTransformer	0.9926	0.7719	0.7098	0.9693	0.5599	0.8967
Residual (2+1)D CNN	0.9107	0.7366	0.6788	0.8651	0.5585	0.8109
3DCNN	0.8654	0.6433	0.5315	0.7690	0.4060	
(2+1)D CNN	0.9878	0.6905	0.5970	0.8499	0.4601	
Frame Baseline	0.9263	0.6813	0.6476	0.7209	0.5879	0.7258
Clip Baseline	0.9653	0.6458	0.5464	0.7548	0.4282	0.7056
Ensemble		0.7653	0.7252	0.9523	0.6214	

A recurring pattern of high precision but low recall indicates missed fakes are the dominant failure mode. Novel generators such as HotShot and Lavie are systematically misclassified as real, while generators visually similar to training sources are detected reliably. Critically, this failure pattern is consistent across all architectures, suggesting models learn source-specific signatures rather than a generalised concept of AI-generated content.

REFERENCES

1. Changjing et al. Jiang. 2025. IVY-Fake: A Large-Scale Dataset for AI-Generated Video Detection. https://huggingface.co/datasets/AI-Safeguard/Ivy-Fake.

IVY-Fake dataset

Training Sources

REAL
- Kinetics-400
- Yonku_1M_10s

FAKE
- DynamicCrafter
- Latte
- OpenSora
- SEINE
- ZeroScope
- Pika

2 real + 6 fake = 8

Test Sources (*UNSEEN*)

REAL
- Kinetics-400-val
- MSR-VTT

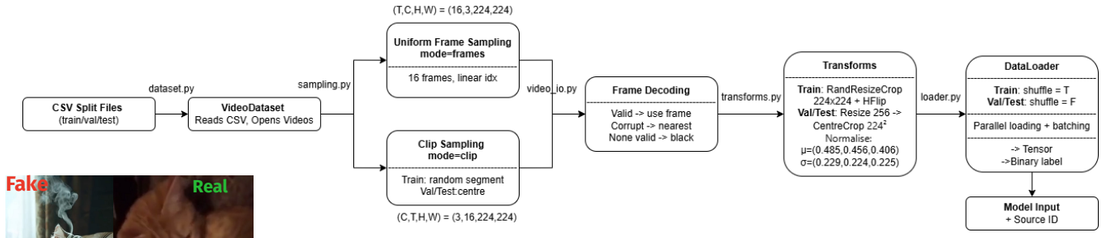
FAKE
- Crafter
- Gen2
- HotShot
- Lavie
- ModelScope
- MoonValley
- MorphStudio
- Sora
- Show_1
- Loki
- WildScape
- Self_craw

2 real + 12 fake = 14

DISCUSSION

The dominant finding is not which model performs best, but that all models fail in the same way a pattern attributed to source-signature learning. Rather than learning a generalised concept of AI-generated content, models learn to recognise specific generators seen during training, causing reliable detection of familiar sources and systematic failure on novel ones. Pretrained backbones offer a modest advantage but do not alter this pattern. The bottleneck is not architectural but a consequence of insufficient training data diversity.

Preprocessing pipeline





INTRODUCTION

Rapid advancements in AI video generation have blurred the visual boundary between real and fake, making robust deepfake detection a societal necessity rather than just an academic curiosity.

Modern AI easily defeats traditional frame-by-frame detection which relies on spatial analysis. The new frontier requires identifying temporal inconsistencies, how physics, lighting, and geometry evolve over time.

To address this, we benchmarked multiple architectures ,from spatial aggregators to explicitly spatiotemporal networks, on the IVY-Fake dataset to evaluate their true, real-world readiness against unseen AI generators.

METHODOLOGY

We benchmarked three distinct architectural families, Clip (3DCNN, Video Swin Transformer and Clip Baseline), Frame (LCRN, and Frame Baseline, and a Trajectory based model ReStraV, on the large-scale IVY-Fake dataset. To guarantee real-world relevance, we implemented a strict open-set testing protocol, evaluating the models exclusively against newly generated AI videos they had never seen during training. All models were trained on a high-performance computing cluster, prioritising strategies that capture true cross-source generalisation rather than simple data memorisation.

CONCLUSION

The LCRN achieves the strongest performance, but the central finding is that source-signature learning is the dominant failure mode across every architecture evaluated independent of design complexity or the use of pretrained weights. The results suggest that the most productive path towards robust open-set detection lies not in refining CNN architectures, but in expanding training data diversity through more comprehensive dataset curation or the adoption of pretrained multimodal backbones with sufficiently broad visual priors.